

Effects of Concurrent Resistance and Endurance Training Using Continuous or Intermittent Protocols on Muscle Hypertrophy: Systematic Review With Meta-Analysis

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Abstract

Monserdà-Vilaró, A, Balsalobre-Fernández, C, Hoffman, JR, Alix-Fages, C, and Jiménez, SL. Effects of concurrent resistance and endurance training using continuous or intermittent protocols on muscle hypertrophy: Systematic review with meta-analysis. *J Strength Cond Res* 37(3): 688–709, 2023—The purpose of this systematic review with meta-analysis was to explore the effects of concurrent resistance and endurance training (CT) incorporating continuous or intermittent endurance training (ET) on whole-muscle and type I and II muscle fiber hypertrophy compared with resistance training (RT) alone. Randomized and nonrandomized studies reporting changes in cross-sectional area at muscle fiber and whole-muscle levels after RT compared with CT were included. Searches for such studies were performed in Web of Science, PubMed, Scopus, SPORTDiscus, and CINAHL electronic databases. The data reported in the included studies were pooled in a random-effects meta-analysis of standardized mean differences (SMDs). Twenty-five studies were included. At the whole-muscle level, there were no significant differences for any comparison ($SMD < 0.03$). By contrast, RT induced greater type I and type II muscle fiber hypertrophy than CT when high-intensity interval training (HIIT) was incorporated alone ($SMD > 0.33$) or combined with continuous ET ($SMD > 0.27$), but not compared with CT incorporating only continuous ET ($SMD < 0.16$). The subgroup analyses of this systematic review and meta-analysis showed that RT induces greater muscle fiber hypertrophy than CT when HIIT is included. However, no CT affected whole-muscle hypertrophy compared with RT.

Key Words: resistance training, strength training, concurrent training, skeletal muscle, hypertrophy, high-intensity interval training

Introduction

Many sports require athletes to perform short-duration, high-intensity efforts (e.g., tackles, sprints, jumps, or throws) while covering long-running distances with minimal rest between actions (58). Therefore, a combination of aerobic-endurance training, anaerobic training, and strength-power training is required to maximize athletic performance (71). To achieve such adaptations, and to protect athletes against injury (44), many coaches integrate these various training regimens within the athlete's periodized training program (i.e., concurrent training). However, when compared with strength training alone, concurrent training may compromise strength and/or hypertrophy adaptations, a phenomenon commonly termed as the "interference effect" (32,46). Although the mechanisms underpinning this interference effect are not well understood, potential candidates include a compromised resistance training stimulus due to residual neuromuscular fatigue, substrate depletion, and/or an attenuated postexercise anabolic response exacerbated by the

increased training volume or potentially through the addition of endurance exercise to the training program (5,24,34).

Indeed, the compatibility of performing concurrent strength and endurance training has been questioned since the classic work of Hickson in 1980 (33), which provided the first evidence of impaired strength adaptations from concurrent training. Since then, many early studies provided robust evidence that gains in strength (7,2), muscle hypertrophy (7,42), and power (26,42) were attenuated by concurrent training, compared with resistance-only training. However, the literature remains inconclusive. Often, "interference" effects are not always reported; some studies have demonstrated similar gains in strength and hypertrophy (17,43,50,51), whereas others have shown that, under certain conditions, concurrent exercise induced greater hypertrophic adaptations than resistance-only training (40,49).

When concurrently training, athletes undergo greater total workloads and might not be able to match the adaptive response (i.e., overreaching) of single-mode training, possibly increasing neuromuscular fatigue and resulting in an exacerbated interference effect (8,14). It has been traditionally hypothesized that the addition of endurance training may result in overreaching or overtraining and is believed to respond in a volume-intensity-dependent manner (14,24,30). In support of this, a meta-analysis by Wilson et al. (78) reported that these 2 training variables may

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Table 1
PICOS question model

P (population)	I (intervention)	C (comparison)	O (outcomes)	S (study design)
Healthy male and female subjects aged ≥ 18 y	Resistance and endurance training performed concurrently	Concurrent training vs the same resistance training program performed alone	Muscle hypertrophy of the whole-muscle and/or myofiber levels	Randomized and nonrandomized comparative studies

have a significant role when mediating any interference effect. Traditional endurance training is typically characterized by high-volume, long-duration contractile activity at a relatively low intensity (4,21). Some studies have reported that high-volume, moderate-intensity, continuous training (MICT) and frequent aerobic exercise may modulate key pathways that downregulate muscle protein synthesis (61), thereby increasing the chance of an interference to muscle hypertrophy gains (20,24,72). However, performing short-duration, high-intensity interval exercise has been shown to result in higher activation of AMPK (11,63,79) (a

protein which is proposed to inhibit mTORC1 (73)) and may therefore exacerbate the potential for an interference effect (13,24). Conversely, other studies have reported that low-volume, high-intensity interval training (HIIT) or maximal or supramaximal sprint interval training (SIT) does not compromise strength and/or whole-muscle hypertrophy adaptations (10,17,29). In support of this, a recent meta-analysis (65) concluded that total body and lower-body lean mass were not negatively affected by performing HIIT with resistance training. However, this study only focused on the effects of concurrent

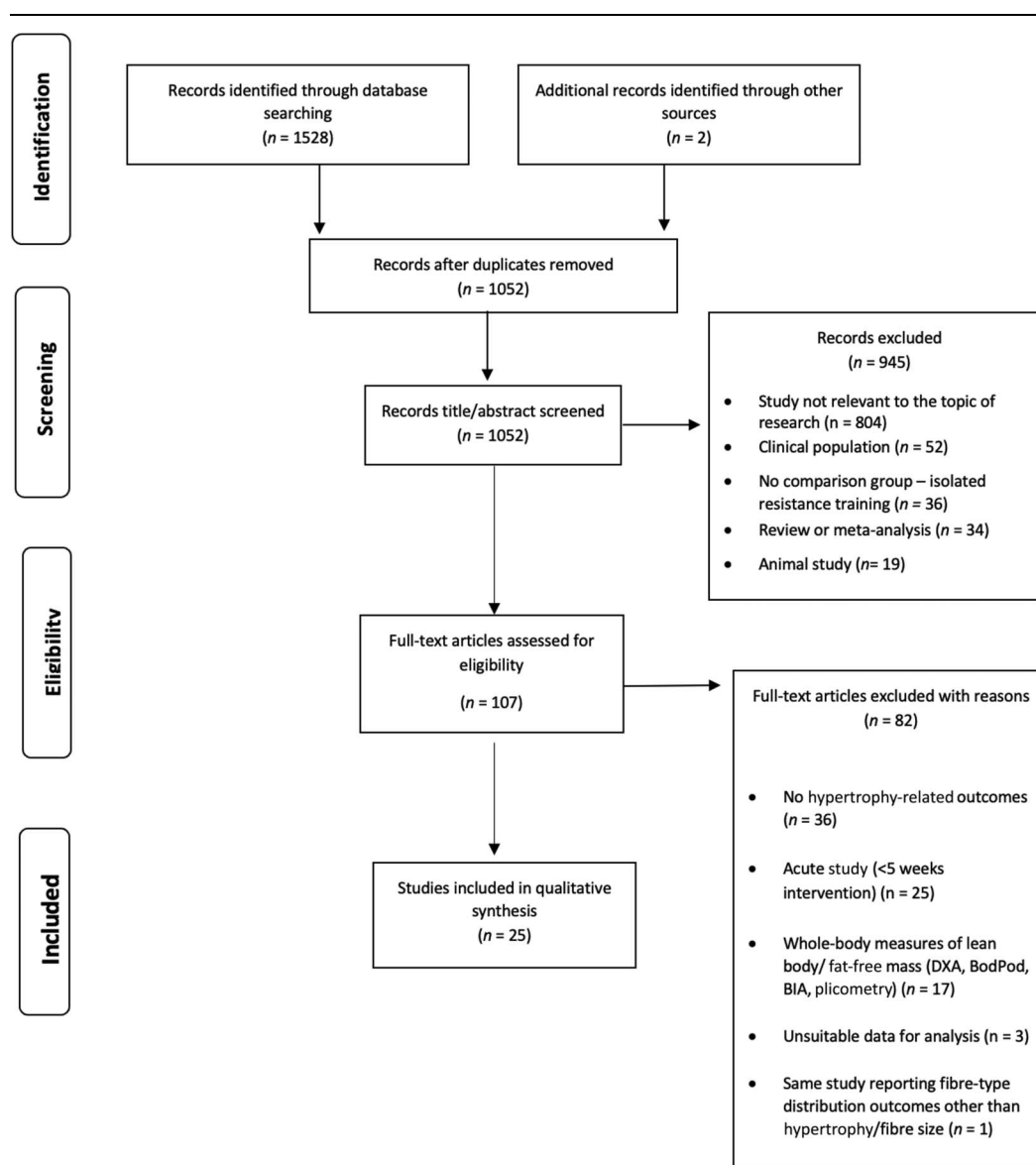
**Figure 1.** PRISMA flowchart for study eligibility.

Table 2

Detailed information on subject characteristics of the included studies.

Study	Group	No of subjects	Sex: M (%)	Age (y)	Height (cm)	Body mass (kg)	Resistance training status	Endurance training status	Study design
Ahtiainen et al. (2)	MXD + RT	7	100	64.0 ± 3.0	173.0 ± 3.0	80.0 ± 7.0	Untrained	Untrained	Randomized
	RT	10	100	61.0 ± 5.0	177.0 ± 3.0	80.0 ± 5.0			
Bell et al. (6)	MICT + RT	16	100	22.0 ± 3.6	178.4 ± 7.2	77.2 ± 9.4	Trained	Trained	Nonrandomized
	RT	15	100	22.0 ± 1.9	178.0 ± 6.3	78.4 ± 8.1			
Bell et al. (7)	MXD + RT	13 (8M 5W)	61.53	22.3 ± 3.3	176.0 ± 9.3	73.4 ± 11.6	Untrained	Untrained	Randomized
	RT	11 (7M 4W)	63.63	22.3 ± 3.3	176.0 ± 9.3	73.4 ± 11.6			
De Souza et al. (17)	HIIT + RT	11	100	22.5 ± 3.9	176.0 ± 8.1	72.9 ± 9.8	Untrained	Untrained	Randomized
	RT	11	100	25.9 ± 6.4	172.0 ± 4.3	73.5 ± 16.1			
De Souza et al. (15)	HIIT + RT	11	100	22.5 ± 3.9	176.0 ± 8.1	72.9 ± 9.8	Untrained	Untrained	Randomized
	RT	11	100	25.9 ± 6.4	172.0 ± 4.3	73.5 ± 16.1			
Fyfe et al. (23)	HIIT + RT	8	100	29.5 ± 2.1	181.3 ± 5.8	82.6 ± 10.9	Trained	Trained	Nonrandomized
	MICT + RT	7	100	30.8 ± 7.1	183.3 ± 4.2	85.5 ± 9.8			
Häkkinen et al. (29)	RT	8	100	28.6 ± 6.4	182.7 ± 7.6	86.6 ± 14	Untrained	Untrained	Nonrandomized
	MXD + RT	11	100	37.0 ± 5.0	181.0 ± 8.0	88.6 ± 12.9			
Irving et al. (35)	RT	16	100	38.0 ± 5.0	179.0 ± 5.0	83.9 ± 15.0	Untrained	Untrained	Randomized
	MICT + RT (young)	12 (6M 6F)	50	26.0 ± 3.5	175.0 ± 6.9	81.0 ± 13.9			
	RT (young)	11 (6M 5F)	54.54	25.0 ± 3.3	170.0 ± 6.6	72.0 ± 9.9	Untrained	Untrained	Randomized
	MICT + RT (older)	10 (5M 5F)	54.54	71.0 ± 6.3	167.0 ± 12.6	80.0 ± 15.8			
	RT (older)	10 (5M 5F)	50	70.0 ± 3.2	169.0 ± 6.3	75.0 ± 15.8	Untrained	Untrained	Randomized
	MXD + RT	30	100	56.0 ± 7.0	177.0 ± 8.0	82.0 ± 13.0			
Karavirta et al. (39)	RT	25	100	56.0 ± 6.0	178.0 ± 6.0	84.0 ± 8.0	Untrained	Untrained	Nonrandomized
	MXD + RT	9	100	26.0 ± 5.3	179.0 ± 9.6	78.0 ± 12.1			
Kazior et al. (40)	RT	7	100	28.0 ± 3.7	182.0 ± 6.0	77.0 ± 6.7	Trained	NR	Randomized
	HIIT + RT	6	100	20.0 ± 1.8	171.2 ± 4.9	64.5 ± 4.7			
Kikuchi et al. (41)	RT	6	100	20.0 ± 1.8	171.2 ± 4.9	64.5 ± 4.7	Trained	Trained	Randomized
	MXD + RT	9	100	23.3 ± 3.6	174.1 ± 6.7	74.2 ± 6.7			
Kraemer et al. (42)	MXD + uRT	9	100	22.9 ± 5.0	176.7 ± 4.0	75.6 ± 8.5	Untrained	Untrained	Nonrandomized
	RT	9	100	24.3 ± 5.1	175.3 ± 6.1	76.6 ± 14.0			
Lundberg et al. (47)	MICT + RT/RT (limb)	10	100	25.0 ± 4.0	184.0 ± 6.0	83.0 ± 13.0	Untrained	Untrained	Nonrandomized
Lundberg et al. (48)	MICT + RT/RT (limb)	10	100	26.0 ± 5.0	183.0 ± 7.0	77.0 ± 9.0	Untrained	Untrained	Nonrandomized
Lundberg et al. (49)	MICT + RT/RT (limb)	10	100	25.0 ± 4.0	184.0 ± 6.0	83.0 ± 13.0	Untrained	Untrained	Nonrandomized
McCarthy et al. (51)	MICT + RT	10	100	27.3 ± 1.7	179.2 ± 1.5	82.1 ± 4.3	Untrained	Untrained	Randomized
	RT	10	100	27.9 ± 1.2	180.4 ± 1.3	82.0 ± 4.4			
Nelson et al. (57)	MICT + RT	5	100	26.0 ± 2.9	NR	89.4 ± 9.1	Untrained	Untrained	Nonrandomized
	RT	5	100	27.0 ± 2.5	NR	69.5 ± 8.50			
Rønnestad et al. (62)	MICT + RT	11	100	27.0 ± 2.0	183 ± 2.0	76 ± 2.8	Untrained	Trained	Nonrandomized
	RT	7	100	26.0 ± 2.0	180 ± 3.0	75.7 ± 3.4			
Sale et al. (66)	HIIT + RT/RT (limb)	8 (4M 4F)	50	20.9 ± 1.4	169.2 ± 13.3	65.7 ± 14.4	Untrained	Untrained	Nonrandomized
Shamim et al. (69)	MXD + RT	12	100	26.0 ± 4.0	177.0 ± 7.0	76.4 ± 10.2	Untrained	Untrained	Nonrandomized
	RT	10	100	24.0 ± 6.0	182.0 ± 8.0	75.5 ± 10.3			
Shamim et al. (68)	MXD + RT	12	100	25.0 ± 5.0	BMI = 24.0 ± 3 kg/m ²		Untrained	Untrained	Nonrandomized
	RT	10	100	25.0 ± 5.0					
Sillanpää et al. (70)	MXD + RT	15	100	56.3 ± 6.8	176.2 ± 9.4	77.0 ± 13.5	Untrained	Untrained	Randomized
	RT	13	100	54.6 ± 6.1	177.3 ± 6.7	79.2 ± 4.7			
Terzis et al. (72)	MICT + RT	10	0	21.6 ± 2.2	166.3 ± 5.4	57.6 ± 6.8	Untrained	NR	Randomized
	RT	10*	0	21.6 ± 2.2	166.3 ± 5.4	57.6 ± 6.8			

Table 2
Detailed information on subject characteristics of the included studies. (Continued)

Study	Group	No of subjects	Sex: M (%)	Age (y)	Height (cm)	Body mass (kg)	Resistance training status	Endurance training status	Study design
Timmins et al. (74)	MXD + RT	12	100	24.6 ± 1.1	179.0 ± 0.0	77.1 ± 2.1	NR	NR	Nonrandomized
	RT	10	100	24.6 ± 1.1	179.0 ± 0.0	77.1 ± 2.1	NR	NR	Nonrandomized
Tsitkanou et al. (75)	HIIT + RT	10	100	21.8 ± 0.6	177.4 ± 1.5	74.2 ± 2.1	Untrained	Untrained	Nonrandomized
	RT	11	100	21.8 ± 0.6	177.4 ± 1.5	74.2 ± 2.1	Untrained	Untrained	Nonrandomized

training incorporating HIIT on whole-muscle hypertrophy. In addition, another recent meta-analysis found no effects of concurrent training compared with resistance training on lower-body whole-muscle hypertrophy (67), but authors did not explore at muscle fiber levels or compare HIIT and MICT when concurrently training.

Although previous meta-analyses explored the effects of concurrent training not only on strength (60) but also on muscle hypertrophy (65,67) adaptations, it is currently unclear whether the endurance training modality (i.e., intermittent vs. continuous) may influence any potential interference to skeletal muscle hypertrophy at both whole-muscle and myofiber levels when performed concomitantly with resistance training. We therefore aimed to systematically review and meta-analyze current evidence on hypertrophic adaptations to resistance training (at the whole-muscle and myofiber levels) compared with concurrent training incorporating continuous or intermittent endurance training and to identify potential reasons for disparities in findings in the literature.

Methods

Literature Search

The Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) recommendations (54) have been followed in this systematic review and meta-analysis. This systematic review with meta-analysis was not previously registered. Searches were conducted from the inception of indexing up to and including November 2021 using the Web of Science, PubMed, Scopus, SPORTDiscus, and CINAHL electronic databases. A combination of the following keywords was used as search terms: ("Concurrent training" OR "Concurrent Strength" OR "Concurrent Endurance" OR "Concurrent exercise" OR "Combined training" OR "Combined exercise" OR "Simultaneous training" OR "Simultaneous exercise") AND ("Resistance training" OR "Resistance exercise" OR "Weight lifting" OR "Strength training") AND ("Hypertrophy" OR "Muscle Thickness" OR "Muscle gains" OR "Cross-sectional Area" OR "CSA" OR "Muscle Hypertrophy" OR "Skeletal Muscle" OR "Muscle growth"). The search results were downloaded and filtered in a spreadsheet (Excel 2016; Microsoft, Washington), where all 1,528 yielded results were referenced for manual duplicate removal. After the removal of any duplicates, a total of 1,052 remaining article titles and abstracts were reviewed. Of the studies initially reviewed, 107 studies were determined to be potentially relevant to the topic based on the information summarized in the abstract. Full-text articles were then assessed for eligibility, and 25 articles were regarded as suitable for inclusion based on the criteria outlined. The reference lists of the articles retrieved in the search were subsequently screened for any additional relevant articles, from which 2 additional records were identified. Three studies (36,37,53) were omitted from analysis because of unsuitable data, leaving 25 final studies for analysis.

Inclusion Criteria

The inclusion and exclusion criteria were defined using the PICOS question model (Table 1). Studies were confined to randomized and nonrandomized comparative studies published in English-language, peer-reviewed journals meeting the following criteria: (a) involved adult subjects (≥ 18 years of age) with no known medical conditions or injuries, (b) interventions of ≥ 5 -week duration that compared at

Table 3

Training characteristics.*

Study	Group	Exercise prescription	End modality	RT modality	Order	Supervision	Frequency	Load progression	Duration
Ahtiainen et al. (2)	MXD + RT	MXD: Weeks 1–7: 30' under the AT $\times$ 2 d $\cdot$ wk $^{-1}$; weeks 7–14: 1st: 45' total (15' under the AT + 10' between the A–AnT + 15' under the AT. 2nd: 60' under the AT; weeks 14–21: 1st: 60' (30' under the AT + 2 $\times$ 10' between the A–AnT + 2 $\times$ 5' above AnT). 2nd: 90' under the AT. RT: Weeks 1–7: 3 $\times$ 15–30RM @ 40–60% 1RM of LP, KE, KF + BP, TPD, LPD, SU, and EF; weeks 7–14: 3 $\times$ 6–12RM @ 60–80% 1RM of LP, KE, KF + BP, TPD, LPD, SU, and EF; weeks 14–21: 3 $\times$ 5–8RM @ 70–90% 1RM of LP, KE, KF + BP, TPD, LPD, SU, and EF	Cycling	FW and MLS	ET and RT on separated days	Supervised	MXD: 2 d $\cdot$ wk $^{-1}$ RT: 1 d $\cdot$ wk $^{-1}$	Yes	21 wk
Bell et al. (6)	RT MICT + RT	Same as the concurrent group above MICT: 40 min @ 85–90% MHR ($\approx$ 75% $\dot{V}O_{2\max}$) increased by 5' every 3 wk until 55'. RT: 2 rounds $\times$ 8–12 reps (30" duration each set @ 1.05 rad $\cdot$ s $^{-1}$) in 12 circuit stations involving KE, KF, HE, SU, LW, and upper-body and lower-body exercises (not specified) RT: Same as the concurrent group above (RT group was allowed to perform 1 weekly moderate-intensity endurance session of a maximum of 30' in an attempt to maintain aerobic fitness.	Rowing	Hydraulic LVR circuits	ET and RT on alternate days	Supervised	MICT: 3 d $\cdot$ wk $^{-1}$ RT: 3 d $\cdot$ wk $^{-1}$	Yes	12 wk
Bell et al. (7)	RT MXD + RT	Same as the concurrent group above MXD: MICT 2 d $\cdot$ wk $^{-1}$ of 30' to 42' (4' increase/week) @ VT ($W \approx$ 173 W); HIIT 1 d $\cdot$ wk $^{-1}$ 4–7 sets $\times$ (3':3' work to rest) @ 90% $\dot{V}O_{2\max}$ ($W \approx$ 291 W). Increases of 1 set every 4 wk. RT: 3 d $\cdot$ wk $^{-1}$ 3–6 sets $\times$ 2–10 reps @ 65–85% 1RM of LP, KE, KF, CR, BP, LPD, SP, and EF. Training load increased every 4 wk.	Cycling	FW and MLS	ET and RT on alternate days	Supervised	MXD: 3d/wk RT: 3 d $\cdot$ wk $^{-1}$	Yes	12 wk
De Souza et al. (17)	RT HIIT + RT	Same as the concurrent group above HIIT: Weeks 1–2: 80% $\dot{V}O_{2\max}$ $\times$ 20 $\times$ 60 s (45" rest between bouts); weeks 3–4: 85–90% $\dot{V}O_{2\max}$ $\times$ 20 $\times$ 60 s (60" rest); weeks 5–6: 95% $\dot{V}O_{2\max}$ $\times$ 20 $\times$ 60 s (60" rest); weeks 7–8: 95–100% $\dot{V}O_{2\max}$ $\times$ 15 $\times$ 60 s (90" rest). RT: Weeks 1–2: 3 $\times$ 12RM (36 total volume); weeks 3–4: 4 $\times$ 8–10RM (32–40 total volume); weeks 5–6: 5 $\times$ 6–8RM (30–40 total volume); weeks 7–8: 3 $\times$ 10–12RM (30–36 total volume) of LP, KE, and KF.	Running	MLS	Same day, balanced ET and RT order (5')	NR	HIIT: 2 d $\cdot$ wk $^{-1}$ RT: 2 d $\cdot$ wk $^{-1}$	Yes	8 wk

Table 3
Training characteristics.* (Continued)

Study	Group	Exercise prescription	End modality	RT modality	Order	Supervision	Frequency	Load progression	Duration
De Souza et al. (15)	RT	RT: Same as the concurrent group above (RT group was allowed to perform 1 weekly moderate-intensity endurance session of a maximum of 30' in an attempt to maintain aerobic fitness)	Running	MLS	Same day, balanced ET and RT order (5')	NR	HIIT: 2 d·wk ⁻¹ RT: 2 d·wk ⁻¹	Yes	8 wk
	HIIT + RT	HIIT: Weeks 1–2: 80% vVo ₂ max × 20 × 60 s (45" rest between bouts); weeks 3–4: 85–90% vVo ₂ max × 20 × 60 s (60" rest); weeks 5–6: 95% vVo ₂ max × 20 × 60 s (60" rest); weeks 7–8: 95–100% vVo ₂ max × 15 × 60 s (90" rest). RT: Weeks 1–2: 3 × 12RM (36 total volume); weeks 3–4: 4 × 8–10RM (32–40 total volume); weeks 5–6: 5 × 6–8RM (30–40 total volume); weeks 7–8: 3 × 10–12RM (30–36 total volume) of LP, KE, and KF.							
Fyfe et al. (23)	RT	RT: Same as the concurrent group above (RT group was allowed to perform 1 weekly moderate-intensity endurance session of a maximum of 30' in an attempt to maintain aerobic fitness)	Cycling	FW and MLS	Same day, ET 10' before RT	Supervised	HIIT: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹	Yes	8 wk
	HIIT + RT	HIIT: Weeks 1–2: 5–7 bouts × 120" @ 120% LT; weeks 3–4: 5–9 bouts × 120" @ 130% LT; weeks 5–6: 7–10 bouts × 120" @ 140% LT; weeks 7–8: 7–11 bouts × 120" @ 150% LT. RT: MON/FRI: LP, BP, SR, KE, and KF. WED: LP, FDBP, LPD, DBLG, and KF; weeks 1–2: 3 × 10–12RM @ 65–70% 1RM. Weeks 3–4: 3 × 6–10RM @ 70–82.5% 1RM. Weeks 5–6: 3–4 × 6–8RM @ 77.5–87.5% 1RM. Weeks 7–8: 3–5 × 4–6RM @ 87.5–90% 1RM.							
	MICT + RT	MICT: Weeks 1–2: 15–21' @ 80% LT; weeks 3–4: 15–24' @ 86.7% LT; weeks 5–6: 21–30' @ 93.3% LT; weeks 7–8: 21–33' @ 100% LT. RT: Same as above					MICT: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹		
Häkkinen et al. (29)	RT MXD + RT	Same as the concurrent groups above MXD: Weeks 1–7: MICT 2 d·wk ⁻¹ 30' below AT; weeks 8–14: 1st: 45' HIIT (15' below AT, 10' between At-AnT, 5' above AnT, 15' under AT) and 2nd: 60' MICT under AT; weeks 14–21: 1st: 60' (15' under the AT, 2 × 10' between AT-AnT, 15 min under the AT) and 2nd: 60–90' under AT. RT: LP, KE, KF, EF, BP, TPD, LPD, SU, LADD, and LABD; weeks 1–7: 3–4 × 10–15RM @	Cycling/ Walking	MLS	ET and RT on separated days	Supervised	MXD: 2 d·wk ⁻¹ RT: 2 d·wk ⁻¹	Yes	21 wk

Table 3

Training characteristics.* (Continued)

Study	Group	Exercise prescription	End modality	RT modality	Order	Supervision	Frequency	Load progression	Duration
Irving et al. (35)	RT	50–70% 1RM. Weeks 8–14: 3–5 × 5–12 RM @ 50–80% 1RM. Weeks 15–21: 3–6 × 3–12RM @ 50–80% 1RM.	Cycling	NR	Same day, ET before RT	Supervised	MICT: 5 d·wk ⁻¹ RT: 4 d·wk ⁻¹	Yes	8 wk
	MICT + RT	Same as the concurrent group above MICT: 30' @ 65% $\dot{V}O_2$ peak 5d/wk RT: 4 × 8–10 4 d·wk ⁻¹ and 2/3 the RT group training volume. Monday: DBLG, LP, KE, KF, and SU; Tuesday: CP, LPD, TE, and BC Thursday: same as Monday; Friday: same as Tuesday							
Karavirta et al. (39)	RT	RT: 4 × 8–10 RM 4 d·wk ⁻¹ Monday: DBLG, LP, KE, KF, TR, and SU; Tuesday: CP, IDBP, LPD, SR, LR, EE, and EF; Thursday: same as Monday; Friday: same as Tuesday	Cycling	NR	ET and RT on separated days	Supervised	MXD: 2d/wk RT: 2 d·wk ⁻¹	Yes	21 wk
	MXD + RT	MXD: Weeks 1–7: 30' below AT 2 d·wk ⁻¹ ; week 5 and 7 included 3 sessions above the AT by doing 10' intervals; weeks 8–14: 1st: 10' interval between AT-AnT, 15' warm-up and 15' cooldown below AT. 2nd: MICT 60' below AT; weeks 15–21: 1st: 2 × 10' intervals between AT-AnT + 2 × 5' above AnT +30' below AT. 2nd: MICT 90' below AT. RT: LP, KE, KF, CR, HADD, BP, EF, TPD, LPD, SU, and SBE exercises were performed. Weeks 1–7: 3 × 12–20RM @ 40–60% 1RM; weeks 8–14: 2–4 × 5–12RM @ 60–80% 1RM; weeks 15–21: 2–4 × 5–8 RM @ 70–85% 1RM + 20% of the LP, KE, and BP performed 5–8RM @ 40–50% 1RM explosively.							
Kazior et al. (40)	RT	Same as the concurrent group above	Cycling	NR	ET and RT on separated days	Supervised	MXD: 2-4d/wk RT: 2-4 d·wk ⁻¹	Yes	7 wk
	MXD + RT	MXD: MICT: Performed at a work rate of 63 ± 1.2% of $\dot{V}O_2$ max (60–65 rpm) with intensity and duration being increased progressively every 2 weeks (session 1: 30' @ 60% $\dot{V}O_2$ max, session 11: 60' @ 60% $\dot{V}O_2$ max). HIIT: Sessions of interval cycling @ 95 ± 1.8 $\dot{V}O_2$ max incorporated on the final 3 wk (sessions 12, 14, 16, 18, and 20) consisting of 6–8 × 2 min @ 95% $\dot{V}O_2$ max. RT: LP started at a load of 70% 1RM and raised 5–7% every third or fourth training session. Volume was undulated and subjects were guided to perform concentric and eccentric phases of 2 s each. Weeks 1–2: 4 × 12; weeks 3–4: 5 × 10; weeks 5–7: 6 × 8; last 3 sessions consisted of 4 × 15, 12, 8, until failure.							

Table 3

Training characteristics.* (Continued)

Study	Group	Exercise prescription	End modality	RT modality	Order	Supervision	Frequency	Load progression	Duration
Kikuchi et al. (41)	RT HIIT + RT	Same as the concurrent group above HIIT: 4 × 30" maximal effort separated in 4' 30" rest intervals; RT: 3 × 10RM @ 80% 1RM of EF (arm curl)	Cycling	MLS	Same day, ET before RT	Supervised	HIIT: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹	Yes	8 wk
Kraemer et al. (42)	RT MXD + RT	RT: Same as the concurrent group above MXD: MICT (MON/THURS): Max distance in 40' @ 80–85% $\dot{V}O_{2\max}$; HIIT (TUES/FRI): 200–800-m intervals @ 95–100+ % $\dot{V}O_{2\max}$. Exercise-to-rest ratio from 1:4 to 1:0.5. RT: MON/THUR: BP, FLY, SP, UR, LPD, SR, EF, SU, KE, KF, CR, and SSQ @ 2–3 × 10–15RM supersetting some exercises; TUE/THURS: BP, SP, EF, LPD, OBT, SU, CR, KE, LP, and DL @ 3–5 × 5–6 RM	Running	FW and MLS	Same day, RT 5–6 h after ET	Supervised	MXD: 4 d·wk ⁻¹ (2 + 2) RT: 4 d·wk ⁻¹	Yes	12 wk
Lundberg et al. (47)	RT MICT + RT	RT: Same as the concurrent group above MICT: 40' one-legged cycling @ 70% of the Wmax at 60 rpm. After 40', workload was increased by ~ 20 W until failure (avg 2' 30") RT: Same trained leg performed 4 × 7 CON-ECC unilateral KE maximal effort repetitions @ ~400 W mean power	Cycling	Fly-wheel ergometer	Same day, RE 6h after ET	Supervised	MICT: 3 d·wk ⁻¹ RT: 2-3 d·wk ⁻¹	NR	5 wk
Lundberg et al. (48)	RT MICT + RT	Same as the concurrent groups above MICT: 40' one-legged cycling @ 70% of the Wmax at 60 rpm. After 40', workload was increased by ~ 20 W until failure (avg 2' 30") RT: Same trained leg performed 4 × 7 CON-ECC unilateral KE maximal effort repetitions @ ~400 W mean power	Cycling	Fly-wheel ergometer	Same day, RE 15' after ET	Supervised	MICT: 3 d·wk ⁻¹ RT: 2-3 d·wk ⁻¹	NR	5 wk
Lundberg et al. (49)	RT MICT + RT	Same as the concurrent groups above MICT: 40' one-legged cycling @ 70% of the Wmax at 60 rpm. After 40', workload was increased by ~ 20 W until failure (avg 2' 30") RT: Same trained leg performed 4 × 7 CON-ECC unilateral KE maximal effort repetitions @ ~400 W mean power	Cycling	Fly-wheel ergometer	Same day, RE 6h after ET	Supervised	MICT: 3 d·wk ⁻¹ RT: 2-3 d·wk ⁻¹	NR	5 wk
McCarthy et al. (51)	RT MICT + RT	Same as the concurrent group above MICT: 50' of continuous cycling ergometry @ 70% heart rate reserve (HRR); RT: 3x 5–7RM of SQ, BP, EF, KE, KF, LPD, SP, and CR.	Cycling	FW and MLS	Same day, order rotated (10')	Supervised	MICT: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹	Yes	10 wk
Nelson et al. (57)	RT MICT + RT	Same as the concurrent group above MICT: Mon/Tues/Thurs/Fri week 1: 30' @ 75% MHR; week 2: 30' @ 80% MHR; week 3: 40' @ 80% MHR; week 4: 50' @ 80% MHR; week 5–6:	Cycling	Isokinetic resistance machine	Same day, RT before ET (10')	Supervised	MICT: 4 d·wk ⁻¹ RT: 4 d·wk ⁻¹	NR	20 wk

Table 3

Training characteristics.* (Continued)

Study	Group	Exercise prescription	End modality	RT modality	Order	Supervision	Frequency	Load progression	Duration
Rønnestad et al. (62)	RT MICT + RT	60' @ 80% MHR; and weeks 7–20: 60' @ 85% MHR. RT: Mon/Tues/Thurs/Fri using an isokinetic device, training consisted of 3 × 6RM of KE and KF @ 30°/sec Same as the concurrent group above MICT: Endurance training consisted of the cyclist preparation phase before competition corresponding to 9.9 ± 1.1 h of training per week. RT: Exercises were SQ, unilateral LP, SLHF, and CR. Weeks 1–3: 1st: 3 × 10RM and 2nd: 3 × 6RM; weeks 4–6: 1st: 3 × 8RM and 2nd: 3 × 5RM; weeks 7–14: 1st: 3 × 6RM and 2nd: 3 × 4RM.	Cycling	FW and MLS	Same day, RT before ET	Supervised	MICT: NR RT: 2 d·wk ⁻¹	Yes	12 wk
Sale et al. (66)	RT HIIT + RT	RT: Same as the concurrent group above HIIT: 3 × 5' bouts @ 90–100% V _{O₂} max. (group A performed unilateral ET and group B performed bilateral ET) RT: 6 × 15–20 RM on the LP (unilateral training) RT: Same as the concurrent group above (group A performed bilateral RT and group B unilateral RT)	Cycling	MLS	Same day, ET before RT	NR	HIIT: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹	NR	22 wk
Shamim et al. (69)	RT MXD + RT	MXD: Endurance training consisted of a mixture of different modes periodized each week: ~40' hill simulator @ 25–100% MAP; moderate-intensity interval training: 8–10 × 2.5' @ 70% MAP, 4–8 × 5' @ 70% MAP; MICT: 45' @ 50% MAP and HIIT at 100% MAP: 3 × 5' @ 70% MAP + 6 × 10" @ 100% MAP + 1 × 60" @ 100% MAP + 3 × 20" @ 100% MAP + 4' @ 40% MAP RT: Exercises performed were whole-body exercises with a focus on LP, KE, and BP. Training ranged between 3 and 5 × 6–15RM @ 60–98% 1RM @ RPE 6–10 depending on the week of training and the number of weekly session.	Cycling	FW and MLS	ET and RT on alternate days	Supervised	MXD: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹	Yes	12 wk
Shamim et al. (68)	RT MXD + RT	Same as the concurrent groups above MXD: Endurance training consisted of a mixture of different modes periodized each week: ~40' hill simulator @ 25–100% MAP; moderate-intensity interval training: 8–10 × 2.5' @ 70% MAP, 4–8 × 5' @ 70% MAP; MICT: 45' @ 50% MAP and HIIT at 100% MAP: 3 × 5' @ 70%	Cycling	FW and MLS	ET and RT on alternate days	Supervised	MXD: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹	Yes	12 wk

Table 3
Training characteristics.* (Continued)

Study	Group	Exercise prescription	End modality	RT modality	Order	Supervision	Frequency	Load progression	Duration
Sillanpää et al. (70)	RT MXD + RT	MAP + 6 × 10'' @ 100% MAP + 1 × 60'' @ 100% MAP + 3 × 20'' @ 100% MAP + 4' @ 40% MAP RT: Exercises performed were whole-body exercises with a focus on LP, KE, and BP. Training ranged between 3 and 5 × 6–15RM @ 60–98% 1RM @ RPE 6–10 depending on the week of training and the number of weekly session. Same as the concurrent groups above MXD: Weeks 1–7: MICT: 30' @ under AT (few sessions above AT); weeks 7–14: 1st: 45' total, 4 loading intervals (15' under AT, 10' between AT-AnT, 5' above AnT, and 15' under AT. 2nd: MICT 60' under AT; weeks 14–21: 1st: 60' total (30' under AT, 2 × 10' between AT-AnT, 2 × 5' above AnT. 2nd: MICT: 90' under AT. RT: LP, KE, KF, BP, TPD, LPD, SU, BE, EF, LABB, and LADD intensities ranged between week cycles. Weeks 1–7: 40–60% 1RM, high number of reps and multiple sets; weeks 7–14: 60–80% 1RM; weeks 14–21: 70–90% 1RM.	Cycling	FW and MLS	ET and RT on different days	Supervised	MXD: 2 d·wk ⁻¹ RT: 2 d·wk ⁻¹	Yes	21 wk
Terzis et al. (72)	RT MICT + RT	Same as the concurrent group above MICT: 30' @ 60–70% MHR (~130–145bpm/ ~9 min/km) velocity gradually increased to ~7.5 min/km during the last week of training. RT: Mon/Fri: 6 × 2 eccentric-only SQ followed by 3 CMJ (weeks 1–2 @ 40% 1RM, weeks 3–4 @ 55% 1RM, weeks 5–6 @ 65% 1RM) Wed: 8 × 3 CMJ + 8 × 3 DJ (weeks 1–2 DJ @ 20 cm, weeks 3–4 DJ @ 30 cm, and weeks 5–6 DJ @ 40 cm)	Running	MLS (power training)	Same day, ET 10' after RT	Supervised	MICT: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹	Yes	6 wk
Timmins et al. (74)	RT MXD + RT	Same as the concurrent group above MXD: Endurance training was made up of a range of simulations including undulating hill efforts (@ 25–110% MAP), MICT (@ 50% MAP), MICT (@ 70% MAP), and HIIT (@ 100% MAP). Moderate-intensity and high-intensity intervals were separated by recovery periods ranging between 20''–60'' @ 40% MAP. RT: RT: 45' LP, RDL, KE, BBLG, BBHT, and SLDL. Undulated training volume depending on the week. Week 1: 5 × 10–15 @ 70% 1RM and RPE 9–10; week 12: 2–3 sets of 2–5 reps at 90–97.5% 1RM and RPE 9–10. Exercises were	Cycling	FW and MLS	ET and RT on alternate days	Supervised	MXD: 3 d·wk ⁻¹ RT: 3 d·wk ⁻¹	Yes	12 wk

Table 3**Training characteristics.*** (Continued)

Study	Group	Exercise prescription	End modality	RT modality	Order	Supervision	Frequency	Load progression	Duration
Tsitkanou et al. (75)	RT	executed with the intent to control the load velocity throughout the whole range of motion.	Cycling	MLS	Same day, ET 10' after RT	NR	HIIT: 2 d·wk ⁻¹ RT: 2 d·wk ⁻¹	Yes	8 wk
	HIIT + RT	Same as the concurrent group above HIIT: 10 × 60'' @ 100% MAP (~183 ± 12 W, 55–60 rev/min) work load was increased by ~2% every training session. RT: 2 × 10 reps of AC, LC, DR + 4 × 6RM @ 80–85% 1RM of LP and HS.							
	RT	Same as the concurrent group above							

*MIX = mixed interval and continuous endurance training; ET = endurance training; CET = continuous endurance training; HIIT = high-intensity interval training; RT = resistance training; AT = aerobic threshold; AnT = anaerobic threshold; RM = repetition maximum; RPE = rate of perceived exertion; LP = leg press; KE = knee extension/quad extension; KF = knee flexion/leg curls; BP = bench press; TPD = triceps push-down; SU = sit-ups/crunches; EF = elbow flexion/biceps curl; LPD = lat pull-down; MHR = maximum heart rate; HE = hip extension; VE = ventilation threshold; SP = shoulder press/military press; CR = calf raises/heel raises/ankle plantar flexion; LT = lactate threshold; SR = seated row; FDBP = flat dumbbell press; DBLG = dumbbell lunges; BBLG = barbell lunges; BBHT = barbell hip thrust; SLDL = stiff-legged deadlift; $\dot{V}O_{2peak}$ = peak oxygen uptake; $\dot{V}O_{2max}$ = $\dot{V}O_{2max}$ velocity; LADD = leg adduction; LABD = leg abduction; TR = toe raises; UR = upright rows; SQ = squat; SSQ = split squat; OBT = oblique twists; DL = deadlift; SLHF = single-leg hip flexion; MAP = maximum aerobic power; BE = back extension; CMJ = counter movement jump; DJ = drop jump.

Table 4

Whole-muscle hypertrophy

Study	Method	Site	Endurance protocol	Endurance modality	Units	ET + RT			RT alone			Main findings	
						n	Pretraining	Posttraining	n	Pretraining	Posttraining	Concurrent training	Resistance training
Ahtiainen et al. (2)	US	VL	MICT + HIIT	Cycling	cm ²	7	2.2 ± 0.3	2.5 ± 0.3	10	2.1 ± 0.3	2.4 ± 0.3	VL muscle thickness: ↑**	VL muscle thickness: ↑***
Bell et al. (6)	CT	VL (QF)	MICT	Rowing	cm ²	14	93.9 ± 12.7	97.7 ± 15.3	14	90.0 ± 8.6	95.1 ± 8.2	KE muscle thickness: ↑*	KE muscle thickness: ↑*
De Souza et al. (17) (L)	MRI	VL	MICT + HIIT	Running	mm ²	11	8340.8 ± 3316.6	8996.8 ± 3049.6	11	8332.4 ± 2962.7	8849.5 ± 2962.7	Left thigh CSA: ↑***	Left thigh CSA: ↑***
De Souza et al. (17) (R)	MRI	VL	MICT + HIIT	Running	mm ²	11	8261.4 ± 3323.3	8882.7 ± 2880.2	11	8215.4 ± 2981.0	8668.0 ± 3158.8	Right thigh CSA: ↑***	Right thigh CSA: ↑***
Häkkinen et al. (29)	MRI	VL	MICT + HIIT	Cycling/Walking	cm ²	8	74.0 ± 16.7	81.6 ± 21.0	10	73.7 ± 18.4	81.4 ± 24.0	QF CSA: ↑***	QF CSA: ↑***
Irving et al. (35) (young)	CT	VL	MICT	Cycling	cm ²	11	259.0 ± 90.1	272.5 ± 15.4	11	259.0 ± 49.8	281.53 ± 28.0	Midthigh CSA: ↑*	Midthigh CSA: ↑*
Irving et al. (35) (old)	CT	VL	MICT	Cycling	cm ²	9	230.0 ± 63.0	241.7 ± 20.3	10	242.0 ± 47.4	255.55 ± 27.5	Midthigh CSA: ↔	Midthigh CSA: ↔
Kikuchi et al. (41)	MRI	BB	HIIT	Cycling	cm ²	6	13.6 ± 1.4	16.3 ± 3.5	6	14.2 ± 2.0	16.6 ± 1.1	BB CSA: ↔	BB CSA: ↑*
Lundberg et al. (47) (mean)	MRI	VL	MICT	Cycling	cm ²	10	79.0 ± 10.0	90.0 ± 10.0	10	78.0 ± 12.0	84.0 ± 12.0	QF CSA: ↑*	QF CSA: ↑*
Lundberg et al. (48)	MRI	VL	MICT	Cycling	cm ²	10	80.7 ± 13.2	85.2 ± 12.9	10	82.2 ± 12.5	84.4 ± 12.5	QF CSA: ↑*	QF CSA: ↑*
McCarthy et al. (51) KE	CT	VL	MICT	Cycling	cm ²	10	89.2 ± 15.9	101.7 ± 14.4	10	93.7 ± 12.3	104.9 ± 14.9	KE CSA: ↑*	KE CSA: ↑*
McCarthy et al. (51) KF	CT	VL	MICT	Cycling	cm ²	10	91.5 ± 20.4	96.82 ± 18.2	10	87.3 ± 16.4	93.5 ± 17.3	KF CSA: ↑*	KF CSA: ↑*
Rønnestad et al. (62) (KE)	MRI	Thigh	MICT	Cycling	cm ²	11	71.4 ± 19.8	80.1 ± 36.5	11	62.0 ± 18.2	80.0 ± 20.8	KE CSA: ↑*	KE CSA: ↑*
Rønnestad et al. (62) (KF)	MRI	Thigh	MICT	Cycling	cm ²	11	52.8 ± 19.7	59.6 ± 20.2	11	51.7 ± 18.5	61.6 ± 20.9	KF CSA: ↑*	KF CSA: ↑*
Sale et al. (66) (KE)	CT	VL	HIIT	Cycling	cm ²	8	35.1 ± 50.1	54.7 ± 64.9	8	30.6 ± 56.0	52.7 ± 60.3	KE CSA: ↑***	KE CSA: ↑***
Sale et al. (66) (KF)	CT	VL	HIIT	Cycling	cm ²	8	53.7 ± 54.4	53.0 ± 42.3	8	48.2 ± 49.1	51.4 ± 43.5	KF CSA: ↔	KF CSA: ↔
Shamim et al. (69)	US	VL	MICT + HIIT	Cycling	cm ³	12	959.0 ± 164.0	1100.0 ± 162.0	10	990.0 ± 122.0	1100.0 ± 134.0	VL muscle volume: ↑*	VL muscle volume: ↑*
Sillanpää et al. (70)	US	TB	MICT + HIIT	Cycling	NR?	15	2.3 ± 0.4	2.7 ± 0.5	13	2.1 ± 0.4	4.3 ± 0.5	TB Thickness: ↑***	TB Thickness: ↑***
Sillanpää et al. (70)	US	VL	MICT + HIIT	Cycling	NR?	15	4.2 ± 0.6	4.5 ± 0.5	13	3.9 ± 0.5	4.3 ± 0.5	VL + VI Thickness: ↑***	VL + VI Thickness: ↑***
Timmins et al. (74)	US	BF	MICT + HIIT	Cycling	cm	12	0.1 ± 0.5	0.2 ± 0.7	10	0.1 ± 0.5	0.2 ± 0.7	BF muscle thickness: ↑* #	BF muscle thickness: ↑*
Tsitkanou et al. (75) (mean*)	US	VL	HIIT	Cycling	cm ²	10	31.2 ± 3.5	38.8 ± 4.7	11	30.3 ± 5.3	38.1 ± 7.0	VL CSA: ↑***	VL CSA: ↑***

Table 5

Myofiber hypertrophy.†

Study	Site	Endurance protocol	Endurance modality	Fiber type	Units	ET + RT		RT alone		Main findings	
						n	Pretraining	n	Pretraining	Concurrent training	Resistance training
Ahtiainen et al. (2)	VL	MICT + HIIT	Cycling	Type I	μm ²	7	5929.0 ± 1249.0	10	5678.0 ± 905.0	Type I CSA: ↔	Type I CSA: ↔
				Type II	μm ²	7	5021.0 ± 829.0	10	4922.0 ± 691.0	Type II CSA: ↔	Type II CSA: ↑*
Bell et al. (7)	VL	MICT + HIIT	Cycling	Type I	μm ²	13	3575.0 ± 862.0	11	3250.0 ± 1423.0	Type I CSA: ↔	Type I CSA: ↑*
				Type II	μm ²	13	3542.0 ± 1053.0	11	3506.0 ± 1529.0	Type II CSA: ↑*	Type II CSA: ↑*
De Souza et al. (15)	VL	HIIT	Running	Type I	μm ²	11	2691.0 ± 340.7	11	2529.0 ± 268.7	Type I CSA: ↔	Type I CSA: ↑*
				Type IIa	μm ²	11	3138.0 ± 552.5	11	3149.0 ± 277.8	Type IIa CSA: ↔	Type IIa CSA: ↑*
				Type IIx	μm ²	11	3062.0 ± 614.9	11	3026.0 ± 212.4	Type IIx CSA: ↔	Type IIx CSA: ↔
Fyfe et al. (23) HIIT	VL	HIIT	Cycling	Type I	μm ²	8	6713.0 ± 1849.0	8	4539.0 ± 848.0	Type I CSA: ↔	Type I CSA: ↑*
				Type II	μm ²	8	6470.0 ± 1481.0	8	5296.0 ± 1347.0	Type II CSA: ↔	Type II CSA: ↔
Fyfe et al. (23) MICT	VL	MICT	Cycling	Type I	μm ²	7	5509.0 ± 2326.0	8	4539.0 ± 848.0	Type I CSA: ↔	Type I CSA: ↔
				Type II	μm ²	7	5051.0 ± 1531.0	8	5296.0 ± 1347.0	Type II CSA: ↔	Type II CSA: ↔
Häkkinen et al. (29)	VL	MICT + HIIT	Cycling/Walking	Type I	μm ²	8	6149.0 ± 2069.0	10	5022.0 ± 1060.0	Type I CSA: ↑*	Type I CSA: ↑*
				Type IIa	μm ²	8	6816.0 ± 1524.0	10	5577.0 ± 1659.0	Type IIa CSA: ↑*	Type IIa CSA: ↑**
				Type IIb	μm ²	8	5660.0 ± 1909.0	10	4836.0 ± 1389.0	Type IIb CSA: ↑**	Type IIb CSA: ↑**
Karavirta et al. (39)	VL	MICT + HIIT	Cycling	Type I	NR?	12	5636.0 ± 1291.0	11	5833.0 ± 951.0	Type I CSA: ↔	Type I CSA: ↔
				Type II	NR?	12	4984.0 ± 1385.0	11	4961.0 ± 1047.0	Type II CSA: ↑**	Type II CSA: ↔
Kazior et al. (40)	VL	MICT + HIIT	Cycling	Type I	μm ²	9	4500.0 ± 1380.0	7	4900.0 ± 435.0	Type I CSA: ↑*	Type I CSA: ↔
				Type IIa	μm ²	9	5680.0 ± 962.0	7	5300.0 ± 875.0	Type IIa CSA: ↑*	Type IIa CSA: ↑*
				Type IIx	μm ²	9	4800.0 ± 1521.0	7	4300.0 ± 288.0	Type IIx CSA: ↑*	Type IIx CSA: ↑*
Kraemer et al. (42) (ST vs C)	VL	MICT + HIIT	Running	Type I	μm ²	9	5088.0 ± 874.0	9	4883.0 ± 1286.0	Type I CSA: ↔	Type I CSA: ↑*
				Type IIa	μm ²	9	5862.0 ± 997.0	9	6084.0 ± 1339.0	Type IIa CSA: ↑*	Type IIa CSA: ↑*
				Type IIb	μm ²	9	5190.0 ± 712.0	9	5795.0 ± 1495.0	Type IIb CSA: ↔	Type IIb CSA: ↔
				Type IIc	μm ²	9	4157.0 ± 983.0	9	3981.0 ± 1535.0	Type IIc CSA: ↔	Type IIc CSA: ↑*
				Type I	μm ²	9	5680.0 ± 535.0	9	4883.0 ± 1286.0	Type I CSA: ↔	Type I CSA: ↑*
Kraemer et al. (42) (ST vs. UC)	VL	MICT + HIIT	Running	Type IIa	μm ²	9	6393.0 ± 1109.0	9	6084.0 ± 1339.0	Type IIa CSA: ↔	Type IIa CSA: ↑*
				Type IIb	μm ²	9	6052.0 ± 1890.0	9	5795.0 ± 1495.0	Type IIb CSA: ↔	Type IIb CSA: ↔
				Type IIc	μm ²	9	3050.0 ± 930.0	9	3981.0 ± 1535.0	Type IIc CSA: ↔	Type IIc CSA: ↑*
Lundberg et al. (47)	VL	MICT	Cycling	Type I	μm ²	10	4055.0 ± 911.0	10	4055.0 ± 911.0	Type I CSA: ↔	Type I CSA: ↔
				Type II	μm ²	10	5046.0 ± 1439.0	10	5046.0 ± 1439.0	Type II CSA: ↑*	Type II CSA: ↔
Lundberg et al. (49)	VL	MICT	Cycling	Type I	μm ²	10	4273.0 ± 821.6	10	4273.0 ± 821.6	Type I CSA: ↑*	Type I CSA: ↔
				Type II	μm ²	10	5302.0 ± 1202.0	10	5302.0 ± 1202.0	Type II CSA: ↑**	Type II CSA: ↑*
McCarthy et al. (51)	VL	MICT + HIIT	Cycling	Type I	μm ²	10	4279.0 ± 452.0	10	4429.0 ± 222.0	Type I CSA: ↔	Type I CSA: ↑**
				Type II	μm ²	10	4422.0 ± 469.0	10	4975.0 ± 399.0	Type II CSA: ↑**	Type II CSA: ↑**
Nelson et al. (57)	VL	MICT	Cycling	Type I	μm ²	5	4638.0 ± 1293.0	5	4813.0 ± 782.6	Type I CSA: ↑*	Type I CSA: ↔
				Type IIa	μm ²	5	6920.0 ± 646.0	5	6502.0 ± 1498.0	Type IIa CSA: ↑*	Type IIa CSA: ↔
				Type IIb	μm ²	5	6006.0 ± 751.0	5	4838.0 ± 731.2	Type IIb CSA: ↑*	Type IIb CSA: ↑*
Shamim et al. (68)	VL	MICT + HIIT	Cycling	Type I	μm ²	12	5733.7 ± 1137.4	10	4631.5 ± 1149.1	Type I CSA: ↔	Type I CSA: ↔
				Type II	μm ²	12	6582.5 ± 1767.7	10	6111.1 ± 1801.3	Type II CSA: ↑*	Type II CSA: ↑*
Terzis et al. (72)	VL	MICT	Running	Type I	μm ²	10	3996.0 ± 933.0	9	3510.6 ± 901.6	Type I CSA: ↔	Type I CSA: ↑*
				Type IIa	μm ²	10	4016.0 ± 1102.0	9	3845.7 ± 861.7	Type IIa CSA: ↔	Type IIa CSA: ↑*
				Type IIx	μm ²	10	2980.4 ± 921.6	9	2856.4 ± 805.8	Type IIx CSA: ↔	Type IIx CSA: ↑*
Tsitkanou et al. (75)	VL	HIIT	Cycling	Type I	μm ²	10	3956.5 ± 959.3	11	4285.7 ± 516.9	Type I CSA: ↑*	Type I CSA: ↑**
				Type IIa	μm ²	10	5599.7 ± 1918.7	11	6006.4 ± 1227.9	Type IIa CSA: ↑**	Type IIa CSA: ↑***
				Type IIx	μm ²	10	4007.0 ± 1439.0	11	4694.8 ± 1550.6	Type IIx CSA: ↑**	Type IIx CSA: ↑**

†ET = endurance training; RT = resistance training; VL = vastus lateralis; MICT = moderate-intensity continuous training; HIIT = high-intensity interval training; ↑ = significant increase from baseline; ↔ = no change from baseline; ↓ = significant decrease from baseline; **p* < 0.05, ***p* < 0.01, ****p* < 0.001, ¹ *n* = 3, ² *n* = 4.

least 1 group performing concurrent training and at least 1 group performing the same resistance training program only, and (c) reported changes in muscle hypertrophy at the whole-muscle level (using ultrasonography, computed tomography, or magnetic resonance imaging) and/or at the muscle fiber level (using histological assessment of muscle biopsies). Studies were excluded if they reported changes in muscle hypertrophy exclusively using dual x-ray absorptiometry, used skinfold measurements, and/or computer image analysis. Studies that used fat-free mass measures to measure skeletal muscle hypertrophy (e.g., plicometry, bioimpedance analysis, or BodPod) were also excluded. Although these techniques are also relevant for assessing body composition, this review focused on more direct muscular assessments to evaluate changes in cross-sectional area at muscle fiber and whole-muscle levels.

Data Extraction

Full-text studies were read, and the relevant data were extracted by one reviewer (A.M.-V.) and checked for accuracy by a second reviewer (S.L.J.). Relevant variables were arranged in 3 different tables and included (a) author names and year of publication, (b) subject characteristics (e.g., age, sex, training status, height, and body mass), (c) specifics of the training interventions (e.g., endurance and resistance training protocols including the number of sets, repetitions, intensity, training frequency, and intervention length), (d) the methods used to assess muscle hypertrophy, and (e) preintervention and postintervention values (as mean $\pm$ SD) for muscle hypertrophy at the whole-muscle or myofiber levels. For studies that reported standard errors, these values were converted to SDs. Attempts were made to contact study authors where relevant data were missing or further clarification was required. Where raw data were not presented in tables or in the text and the authors could not be reached, data were extracted using WebPlotDigitizer (WebPlotDigitizer, V.4.2. Texas: Ankit Rohatgi, 2021) or calculated from baseline values and/or percentage change. Where the required data were not obtainable, the study was excluded from the meta-analysis.

Methodological Quality

The quality of each study was independently assessed by 2 of the authors (A.M.-V. and S.L.J.), and an agreement was mutually determined for any observed discrepancies. Disagreements between ratings were resolved by discussion; if no consensus was reached by discussion, a third reviewer was sought (C.B.-F.). The methodological quality of studies meeting the inclusion criteria was assessed using a modified Downs and Black quality assessment tool (18), which is appropriate for the evaluation of the methodological quality of randomized and nonrandomized studies of health care interventions (27,28). In the modified checklist used, the maximum score was 29 points, with higher scores reflecting higher-quality studies. Scores above 20 were considered of good methodological quality, scores of 11–20 were considered of moderate methodological quality, and scores below 11 were considered of poor methodological quality.

Statistical Analyses

Standardized mean difference (SMD) with 95% confidence intervals (CIs) between different concurrent training modalities and resistance training was calculated using a random-effects model. Main analyses were performed to explore the effects of concurrent training on whole-muscle and type I and type II

muscle fiber hypertrophy compared with resistance training alone. Subgroup analyses were performed to explore how concurrent training incorporating high-intensity interval training, continuous endurance training, or both combined influences whole-muscle and type I and type II muscle fiber hypertrophy compared with resistance training alone. Mean and SDs of the preintervention and postintervention changes for each training modality were obtained from original articles. Heterogeneity of the analyzed studies was assessed using an I² test, setting the significance level at $p < 0.01$. A score below 40% was interpreted as no heterogeneity, whereas scores between 40 and 60% was interpreted as moderate heterogeneity following Cochrane's recommendations. The following threshold values were used to interpret SMD: <0.25 , trivial; 0.25 – 0.50 , small; 0.50 – 1.0 , moderate; and >1.0 , large. The level of statistical significance was set at $p < 0.05$, and all calculations were performed with RevMan 5.4.0 for Mac OS.

Results

Search Results

The original search yielded 1,528 results, including 359 from Web of Science, 226 from Scopus, 234 from PubMed, 633 from CINAHL, and 76 from SPORTDiscus. Two studies were added after screening through the reference lists of the final articles. After removing duplicates, 1,052 articles were identified in the electronic database search from which 945 records were excluded. One hundred seven articles were determined to be potentially relevant based on the information in the abstract and were then fully read for assessment of eligibility. After full-text reading, 82 articles were excluded. The remaining 25 articles were included in the meta-analysis (Figure 1).

Study Characteristics

Subjects' Characteristics. When combined, a total sample of 540 individuals (486 men and 54 women) participated in the included studies. Subjects were all healthy adults ranging between 20- and 71-year-old (31.0 ± 3.8 years). Subject's training status was classified based on whether subjects were reported to have recent experience (e.g., previous 6 months) in endurance and/or resistance training; hence, the level of experience (untrained or trained) is reported for all studies. Of the 25 articles, 18 involved untrained subjects (2,7,15,17,29,35,39,40,47–49,51,57,66,68–70,75), 3 studies included trained subjects (6,23,42), 1 article included resistance-trained subjects but did not report the endurance training status (41), 1 article included endurance-trained subjects with no RT experience (62), 1 article involved resistance-untrained populations but did not report any data regarding the endurance training experience (72), and 1 study did not provide any information concerning the training experience (74). Table 2 provides more detailed information on subject characteristics of the included studies.

Training Characteristics. Training interventions ranged from 5 to 22 weeks with a mean duration of 12 weeks. The most common training frequency was 3 times per week (range 2–5 times per week). Full-body (FB) training and lower-body (LB) training were the most common training splits, with 10 studies (2,6,7,23,29,39,42,51,68–70) implementing a full-body routine. Seven other studies (2,6,7,23,29,39,42,51,68–70) prescribed

Table 6
Results of the methodological quality assessment using the modified Downs and Black checklist.*

Study	Items	Reporting ^a										External Validity ^b			Bias ^c							Confounding ^d										Score Total/ 29
		01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29		
Ahtiainen et al. (2)	1	1	1	1	1	1	1	1	1	1	0	1	0 ^e	1	0 ^e	0 ^e	1	1	1	0 ^e	1	1	0 ^e	1	0 ^e	1	1	1	0	1	21-G	
Bell et al. (6)	1	1	1	1	1	0	1	1	1	1	0	0 ^e	0 ^e	1	0 ^e	0 ^e	1	1	1	0 ^e	1	1	0 ^e	0	0 ^e	0	1	0	0	0 ^e	15-M	
Bell et al. (7)	1	1	1	1	1	0	1	1	1	1	0	0 ^e	0 ^e	1	0 ^e	0 ^e	1	1	1	0 ^e	1	0 ^e	0 ^e	1	0 ^e	0	1	1	0	0 ^e	16-M	
De Souza et al. (17)	1	1	1	1	1	1	1	1	0	0	0	1	1	1	0 ^e	1	1	1	1	0 ^e	1	1	0 ^e	1	0 ^e	0	1	1	0	0 ^e	19-M	
De Souza et al. (15)	1	1	1	1	1	1	1	1	0	0	0	1	1	1	0 ^e	0 ^e	1	1	1	0 ^e	1	1	0 ^e	1	0 ^e	0	1	1	0	0 ^e	18-M	
Fyfe et al. (23)	1	1	1	1	1	1	1	1	1	1	1	0 ^e	1	1	0 ^e	1	1	1	1	0 ^e	1	0 ^e	0 ^e	1	0 ^e	1	1	1	1	0	0 ^e	21-G
Häkkinen et al. (29)	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0 ^e	0 ^e	1	1	1	0 ^e	1	1	0 ^e	0	0 ^e	1	1	0 ^e	0 ^e	1	20-G	
Irving et al. (35)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0 ^e	1	1	1	0 ^e	1	1	0 ^e	1	0 ^e	1	1	1	1	0 ^e	23-G	
Karavirta et al. (39)	1	1	1	1	1	0	1	1	0	0	1	1	0	1	0 ^e	0 ^e	1	1	1	0 ^e	1	1	0 ^e	1	0 ^e	0	1	1	0	1	18-M	
Kazior et al. (40)	1	1	1	1	1	0	1	1	1	0	1	0 ^e	0 ^e	1	0 ^e	0 ^e	1	1	1	0 ^e	1	0 ^e	0 ^e	0 ^e	0 ^e	1	1	0	0 ^e	1	16-M	
Kikuchi et al. (41)	1	1	1	1	1	1	1	1	0	1	1	0 ^e	0 ^e	1	0	0	1	1	1	0 ^e	1	0 ^e	0 ^e	1	0 ^e	0 ^e	1	1	0	1	18-M	
Kraemer et al. (42)	1	1	1	1	1	1	1	1	0	0	0	1	0	1	0 ^e	0 ^e	1	1	1	1	1	1	0 ^e	1	0 ^e	1	1	1	1	1	21-G	
Lundberg et al. (47)	1	1	1	1	0	1	1	1	1	1	1	1	0 ^e	1	0	1	1	1	1	1	1	1	0 ^e	1	0 ^e	1	1	0 ^e	1	1	23-G	
Lundberg et al. (48)	1	1	1	1	1	0	1	1	1	1	1	1	1	1	0	0	1	1	1	0 ^e	1	1	0 ^e	1	0 ^e	1	1	0 ^e	0	1	21-G	
Lundberg et al. (49)	1	1	1	1	1	0	1	1	1	1	1	1	0 ^e	1	0	0	1	1	1	0 ^e	1	1	0 ^e	0	0 ^e	0 ^e	1	1	0	1	19-M	
McCarthy et al. (51)	1	1	1	1	1	0	1	1	0	1	1	0	0	1	0 ^e	1	1	1	1	1	1	0 ^e	0 ^e	1	0 ^e	0	1	1	1	1	20-G	
Nelson et al. (57)	1	1	1	1	1	0	1	1	0	0	0	0	0	1	0 ^e	0 ^e	1	1	1	0 ^e	1	0 ^e	0 ^e	1	0 ^e	0	1	1	0	1	15-M	
Rønnestad et al. (62)	1	1	1	1	1	0	1	1	1	1	1	0 ^e	0 ^e	1	0	0 ^e	1	1	1	1	1	0 ^e	0 ^e	0	0 ^e	0 ^e	1	1	1	1	19-M	
Sale et al. (66)	1	1	1	1	1	0	1	1	1	1	1	0 ^e	0 ^e	1	0	1	1	1	1	0 ^e	1	0 ^e	0 ^e	1	0 ^e	0 ^e	1	1	0	0 ^e	18-M	
Shamim et al. (69)	1	1	1	1	1	0	1	1	1	1	1	0 ^e	1	1	0	0	1	1	1	0 ^e	1	0 ^e	0 ^e	0 ^e	0 ^e	0 ^e	1	1	0	1	18-M	
Shamim et al. (68)	1	1	1	1	1	1	1	1	1	1	1	0 ^e	1	1	0	0	1	1	1	0 ^e	1	0 ^e	0 ^e	0 ^e	0 ^e	0 ^e	1	1	0	1	19-M	
Sillanpää et al. (70)	1	1	1	1	1	0	1	1	0	0	1	1	0	1	0 ^e	0 ^e	1	1	1	1	1	1	0 ^e	1	0 ^e	0	1	1	1	1	20-G	
Terzis et al. (72)	1	1	1	1	1	0	1	1	1	1	1	0 ^e	0 ^e	1	0	0	1	1	1	1	1	1	0 ^e	1	0 ^e	0 ^e	1	1	1	1	21-G	
Timmins et al. (74)	1	1	1	1	1	0	1	1	1	1	1	0 ^e	1	1	0	1	1	1	1	1	1	0 ^e	0 ^e	0 ^e	0 ^e	1	1	1	1	1	22-G	
Tsitkanou et al. (75)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0 ^e	1	1	1	0 ^e	1	1	0 ^e	0	0 ^e	0 ^e	1	1	0	1	21-G	

*1 = criteria met, 0 = criteria not met 0^e = item was unable to be determined, scored 0. ^a Reporting category includes items such as study aims, reported outcomes, subjects characteristics, confounders, adverse events and loss to follow-up; ^b External validity includes questions regarding the study population; ^c Internal validity: bias includes items such as blinding, follow-up and compliance; ^d Internal validity: confounding includes items such as study selection, randomization, and study power.

only lower-body exercises by implementing a LB split. Five studies (40,47–49,66) prescribed only machine leg presses or knee extensions for the lower body; similarly, 1 study prescribed only 3 sets of elbow flexion for the upper body (41). One study distributed exercises between the upper and lower body, training each body part on alternate days (35). The resistance training intensity ranged between 60 and 90% of the subjects 1 repetition maximum (1RM) with most of the working sets and repetitions between 3–6 and 3–12RM, respectively. One study's (72) training

intervention aimed to induce leg muscle power adaptations in preference to hypertrophy. The prescribed power training included 6 sets of 2 repetitions at 55–65% of the 1RM of eccentric-only half squats, followed by CMJ and drop jumps. Five of the 22 studies performed high-intensity interval training (15,17,41,66,75), 9 studies (6,35,47–49,51,57,62,72) used MICT, and 8 studies used a mixed (MXD) endurance training protocol by combining both intermittent and continuous endurance training regimens throughout the length of the intervention

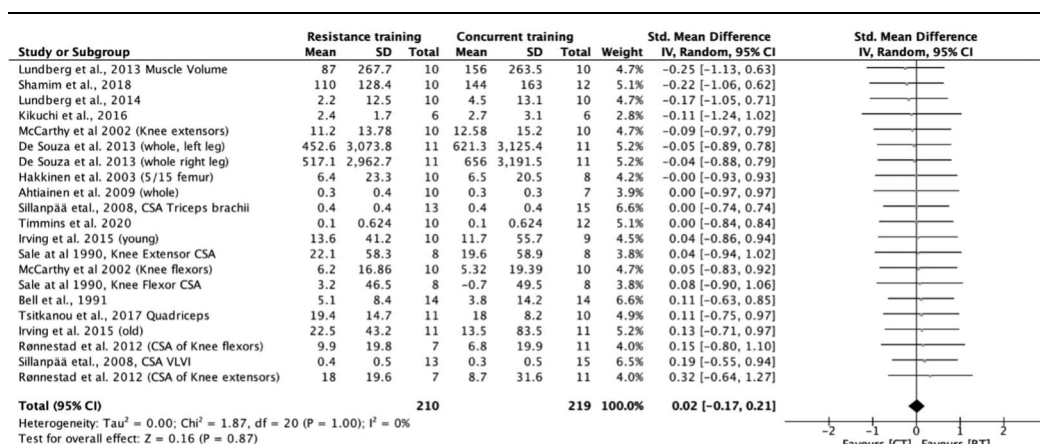


Figure 2. Whole-muscle hypertrophy when considering all studies.

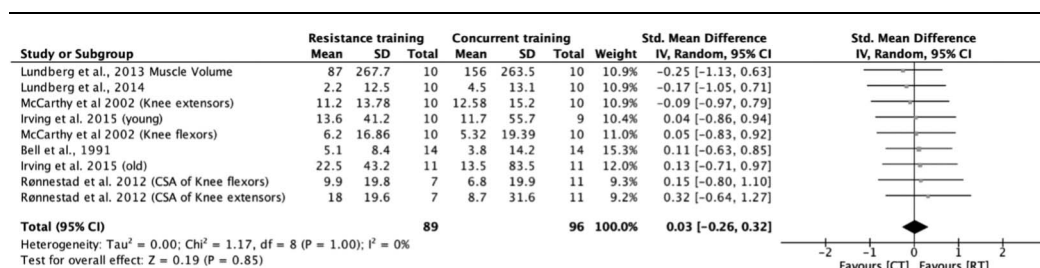


Figure 3. Whole-muscle hypertrophy with continuous concurrent training.

(2,7,29,39,40,42,68–70,74). One study (20) isolated 3 groups performing MICT + RT, HIIT + RT, or RT only. Most of the studies used cycling for the endurance training, whereas 4 studies used running (15,17,42,72), and 1 study used continuous rowing (6). A summary of the training programs and study details from each individual study is provided in Table 3.

Hypertrophy Outcomes. Assessments of muscular hypertrophy were reported at either the whole-muscle (Table 4) or myofiber levels (Table 5). Although most of the studies reported muscle hypertrophy at either the whole-muscle [muscle cross-sectional area (CSA)] or muscle fiber (i.e., muscle fiber (fCSA) levels), 6 studies reported results at both levels (2,29,47,48,51,75). Six studies assessed whole-muscle hypertrophy using magnetic resonance imaging (17,29,41,47,48,62), 4 studies used computed tomography (6,35,51,66), and 5 studies used ultrasound (2,69,70,74,75). For the assessment of muscle fiber size, most studies analyzed muscle biopsies from the vastus lateralis using histochemical staining assays for myofibrillar ATPase activity. Four studies (23,47,49,68) used immunohistochemistry techniques to identify fiber types. Only 1 study (41) used samples from the biceps brachii, and 1 study (70) used samples from both the vastus lateralis and the triceps brachii.

Methodological Quality

Based on the Downs and Black checklist, all studies were considered as of good or moderate methodological quality. Of possible score of 29, the mean $\pm$ SD score on the checklist was 19.3 ± 2.2 (range 15–23). Twelve studies (2,23,29,35,42,47,48,51,70,72,74,75) were classified as being of good quality, whereas the remaining 13 studies were classified as being of moderate quality. All studies scored 0 (i.e., not reported or unable to determine) for reporting whether subjects were representative of the population from which they were recruited and for using a randomized intervention from both subjects and researchers. All studies reported the aims and purpose of the study, the main outcomes, included a description of the subjects' characteristics, described the interventions, reported the main findings, provided estimates of the random variability, included staff and

facilities representative of the intervention the subjects received, ensured the time between the intervention and outcome was approximately the same for all subjects, and provided accurate and reliable outcome measures. Only 7 studies reported subject's compliance with the training programs with the range being 92–100%, thus received a point on the item 28 (42,47,50,62,70,72,74). Eight studies did not randomize subjects to intervention groups (6,29,40,49,62,69,74,75). Only 6 studies blinded the researchers measuring the main outcomes (17,23,47,50,66,74). The methodological quality ratings for all studies can be found in Table 6.

Meta-Analysis Results

Whole-Muscle Hypertrophy. When considering whole-muscle hypertrophy, there were no differences between RT and concurrent training incorporating all forms of endurance training (ES = 0.02; 95% CI = -0.17, 0.21; $p = 0.87$) (Figure 2), whereas subanalyses revealed no effect of incorporating either HIIT or MICT endurance training modalities or a combination of both (Figures 3–5). No heterogeneity between studies was observed ($I^2 = 0\%$).

Type I Fiber Hypertrophy. When considering all forms of endurance training incorporated into the concurrent training programs, there seemed to be a benefit of resistance training for type I muscle fiber hypertrophy (ES: 0.30; 95% CI: 0.02, 0.57; $I^2 = 30\%$; $p = 0.03$) (Figure 6).

Subgroup analyses also revealed a moderate significant advantage of resistance training only for stimulating type I muscle fiber hypertrophy in studies using only HIIT (ES: 0.57; 95% CI: 0.04, 1.10; $I^2 = 0\%$; $p = 0.03$) (Figure 7) as the endurance training modality. When using a combination of HIIT and continuous training, a small but nonsignificant advantage of resistance training only was found (ES: 0.27; 95% CI: -0.12, 0.67; $I^2 = 31\%$; $p = 0.17$) (Figure 8). In the subgroup analysis for studies that used only continuous aerobic training, there was no difference between resistance training and concurrent training for type I muscle fiber hypertrophy (ES: 0.16; 95% CI: -0.39, 0.71; $I^2 = 46\%$; $p = 0.56$) (Figure 9).

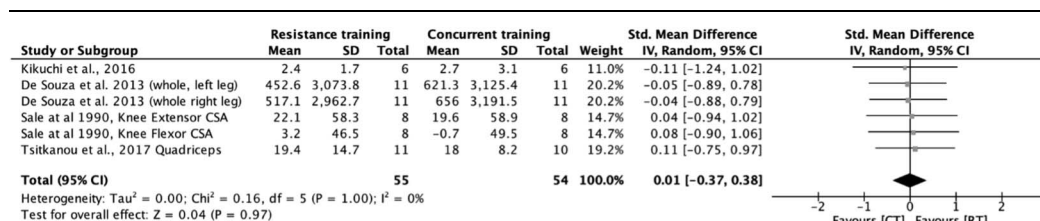


Figure 4. Whole-muscle hypertrophy with intermittent concurrent training.

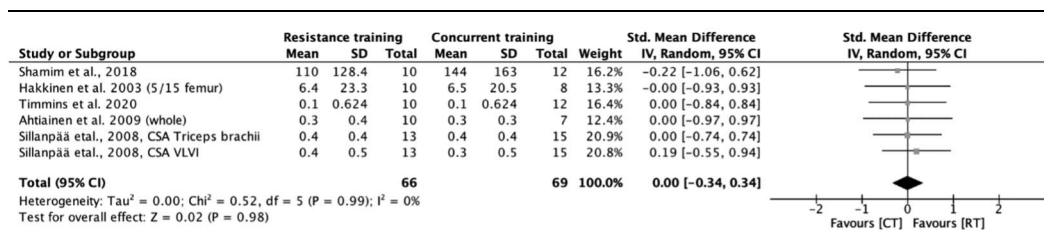


Figure 5. Whole-muscle hypertrophy with mixed concurrent training.

Type II Fiber Hypertrophy. Trivial but significant differences in type II fiber hypertrophy were observed favoring resistance training over concurrent training when considering all studies (ES: 0.23; 95% CI: 0.02, 0.43; $I^2 = 0\%$; $p = 0.03$) (Figure 10).

Subgroup analyses also revealed a small significant advantage of resistance training only for stimulating type II muscle fiber hypertrophy in studies using a combination of HIIT and continuous training (ES: 0.35; 95% CI: 0.01, 0.68; $I^2 = 0\%$; $p = 0.04$) (Figure 11) as the endurance training modality. When using only HIIT, a small but nonsignificant advantage of resistance training only was found (ES: 0.33; 95% CI: -0.11, 0.77; $I^2 = 0\%$; $p = 0.15$) (Figure 12). In the subgroup analysis for studies that used only continuous aerobic training, there was no difference between resistance training and concurrent training for type II muscle fiber hypertrophy (ES: 0.06; 95% CI: -0.27, 0.38; $I^2 = 0\%$; $p = 0.73$) (Figure 13).

Discussion

This is the first meta-analysis to specifically address how concurrent training incorporating different training modalities (high-intensity interval training vs. continuous endurance training) influences both whole-muscle and muscle fiber hypertrophy compared with resistance training alone. At the whole-muscle level, there were no significant differences for hypertrophy outcomes between resistance training and concurrent training, regardless of the endurance training modality used. By contrast, the findings suggest that resistance training induces greater type I and type II muscle fiber hypertrophy than concurrent training incorporating either high-intensity interval training alone or in combination with continuous endurance training, but not compared with concurrent training incorporating continuous endurance training. Although the pooled effects were trivial-to-moderate, some of the 95% confidence intervals were wide, suggesting that the true

effect in the population might even be in the range of “very large” effect. This study strengthens the growing body of evidence that aerobic exercise does not interfere with the hypertrophic potential of resistance training at the whole-muscle level, but type I and type II muscle fiber hypertrophy could be impaired when incorporating high-intensity interval training.

Although concurrent training did not affect whole-muscle hypertrophy compared with resistance training, type I and type II muscle fiber hypertrophy was impaired by concurrent training. In this regard, although both whole-muscle CSA and fCSA measurements represent skeletal muscle hypertrophy, several studies have reported different increases in whole-muscle CSA compared with fCSA in response to resistance training (1,21,76). Note that water compose around 75% of skeletal muscle tissue, whereas intracellular (i.e., beneath the muscle fiber membrane) protein is around 15%; being intracellular, protein was composed 60–70% by myofibrillar proteins, 20–30% by sarcoplasmic proteins, and 10% by mitochondrial proteins (31). Besides, after the first week of resistance training, muscle fiber growth seems to be mainly due to sarcoplasmic hypertrophy, which, based on the sarcoplasm expansion hypothesis of the hypertrophy model, could hypothetically prepare muscle cells for the subsequent myofibrillar hypertrophy (61). Knowing that most of the included studies contained untrained subjects (detailed below) and that a reduction of other muscle components could accompany the sarcoplasmic hypertrophy in these early phases of resistance training (61), not only the different techniques for assessing whole-muscle CSA and fCSA but also the different tissue composition could affect the different results. On the other hand, the higher increases in fCSA in response to resistance training compared with whole-muscle CSA could hypothetically make fCSA more sensitive for the possible interference effect induced by concurrent training because the effect size of the training adaptations is larger and

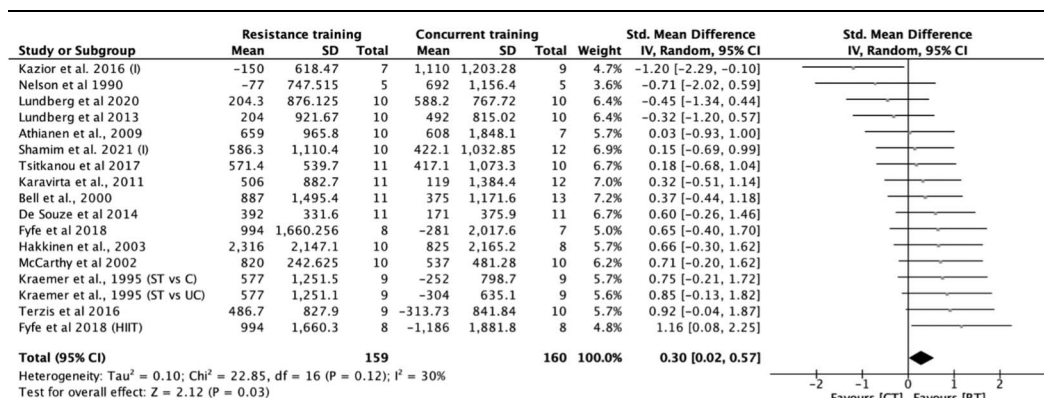


Figure 6. Type I muscle fiber hypertrophy when considering all studies.

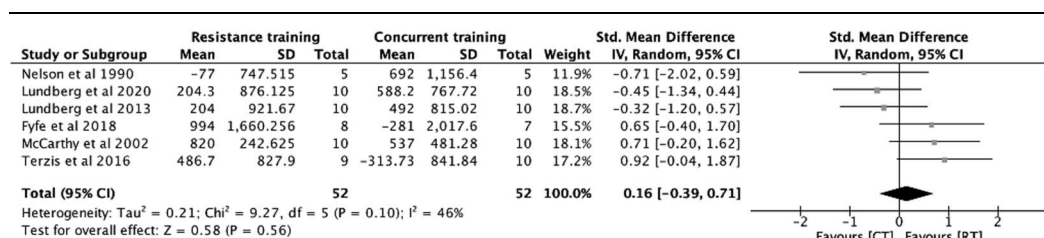


Figure 7. Type I muscle fiber hypertrophy with intermittent concurrent training.

differences between protocols could be clearer. Furthermore, based on the sarcoplasm expansion hypothesis of the hypertrophy model explained above, if concurrent training limits the early sarcoplasmic hypertrophy, long-term skeletal muscle hypertrophy could also be affected.

An interesting finding of this work is that, although concurrent training incorporating high-intensity interval training alone or in combination with continuous endurance training impaired type I and type II muscle fiber hypertrophy compared with resistance training, concurrent training incorporating only continuous endurance training did not affect hypertrophy. These results reject the common proposal of including short-duration high-intensity endurance training instead of low-intensity long-duration exercise to avoid the possible interference effect induced by concurrent training (78). In this sense, our findings do not support the molecular hypothesis defending that it is recommendable to include short-duration high-intensity endurance exercise with concurrent training to keep the activation of AMPK low and favor the AKT-mTOR pathway activation (52). In fact, endurance training performed at higher intensities induces larger AMPK activation (19); thus, lower-intensity continuous endurance training may be more beneficial to keep the activation of AMPK low and favor the AKT-mTOR pathway activation. Besides, note that short-duration high-intensity endurance exercise has demonstrated to induce not only peripheral (i.e., fatigue attributed to processes at or distal to the neuromuscular junction) but also central fatigue (i.e., fatigue attributed to processes within the nervous system), impairing the ability of the central nervous system to fully recruit the muscles to produce tension and force (26,59). Knowing the key role of the neural stimulus (3) and muscle tension on skeletal muscle hypertrophy (77), the neuromuscular fatigue induced by short-duration high-intensity endurance exercise could impair the muscle fiber hypertrophy induced by resistance training if performed under residual fatigue. After a bout of endurance exercise, the force production of the exercise musculature is reduced for at least 6 hours, returning to baseline by 24 hours postexercise with high-intensity endurance training being more fatiguing and leading to larger acute decreases in strength compared with volume-equated moderate continuous training (16,22,38,52,55). Moreover, residual fatigue from previous endurance bouts also

reduces the volume of work performed during upcoming RT sessions, thus limiting any potential for skeletal muscle hypertrophy (12,24,55).

It is important to keep timing and sequence in mind because in 8 of the studies (23,35,41,42,47–49,66) included in this meta-analysis, resistance training took place after endurance training. Nevertheless, some of them separated both trainings by 5–6 hours (42,47,49). A recent systematic review and meta-analysis showed that impairments in strength gains seem to be present only when resistance and endurance training are performed within a short interval between each other (<20 minutes), i.e., within the same training session but not when performed separately (>2 hours) (60). Lee et al. (45) reported that a 3-hour between-mode recovery window was enough to not hinder maximal strength and DXA-derived lean mass gains when performing HIIT + RT compared with RT alone regardless of the order of succession between modes. This 3-hour duration recovery altogether with nutritional strategies (20 g protein after every workout) to support subsequent exercise performance and training adaptation may have afforded the moderately trained subjects an adequate scenario for recovery and preventing an interference effect in hypertrophy during 9 weeks. The same authors also reported that exercise order may alter the subjects “readiness” to train by negatively affecting their subjective perceptions of effort during RT sessions when performed after HIIT sessions which suggest that the physiological stress associated with a previous bout of HIIT (e.g., residual fatigue) had not fully dissipated after the 3-hour recovery. Although Lee et al. (45) used moderately trained individuals, results may be different in highly trained specialized individuals, especially those undertaking more complex training structures involving longer periodized approaches. In other words, athletes performing concurrent training should choose exercise order based on individual preference or the goals of a specific training phase or block wherein doses of different training types and training variables are manipulated to elicit specific peaks in concrete characteristics (e.g., block periodization or phase potentiation), warranting a cautioned interpretation of expected changes in hypertrophy, power, or strength when considering different training approaches. In the same light, a recent meta-analysis (56) suggested that if the primary goal of the cycle is

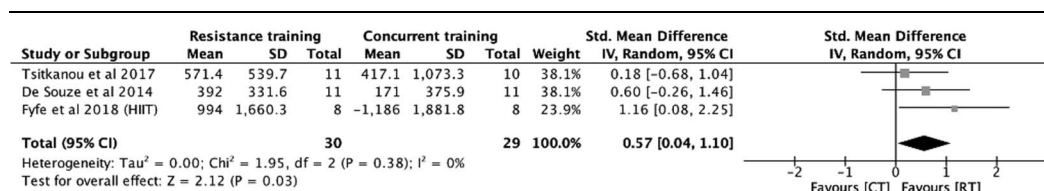


Figure 8. Type I muscle fiber hypertrophy with mixed concurrent training.

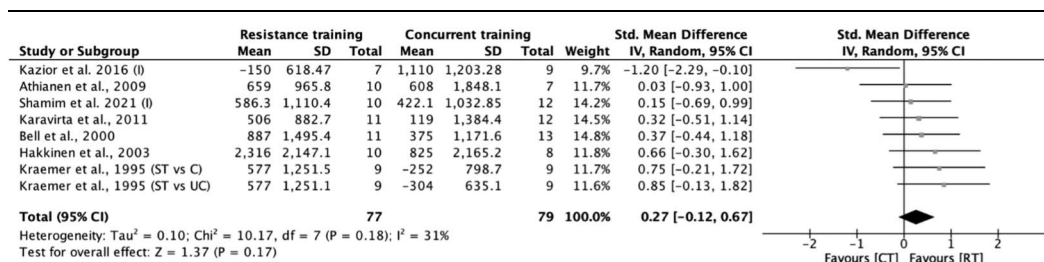


Figure 9. Type I muscle fiber hypertrophy with continuous concurrent training.

to increase body muscular strength, it is highly recommended to perform resistance exercise before endurance training because placing resistance training before endurance training resulted, on average, in a 4-kg higher squat 1RM after a training intervention when compared with the group performing ET before RT. It would be coherent to think then that acute diminished strength performance could affect the progress of chronic adaptations to hypertrophy.

Training status is a major concern when studying hypertrophy adaptations with concurrent training. Although this work did not aim to compare trained and untrained individuals, it should be noted that for type I and type II muscle fiber meta-analyses, only 2 studies included resistance-trained subjects (23,42). In addition, only 2 studies from whole-muscle meta-analyses included resistance-trained subjects (6,41). In this sense, it has been demonstrated that the development of the maximal dynamic strength during concurrent training is only impaired in resistance-trained individuals (60). It should be noted that authors considered subjects as trained when they were athletes or individuals who participated in regular structured training programs for at least 3 months before the intervention period. Besides, some moderately trained individuals (i.e., individuals classified as recreationally or physically active but not involved in a regular structured training program for at least 3 months before the intervention period) also experience negative effects of concurrent training (60). Note that the 2 studies of the fCSA meta-analyses that included resistance-trained subjects found detrimental effects of concurrent training (23,42). Further research is needed to explore whether trained individuals are negatively affected by concurrent training when skeletal muscle hypertrophy is targeted.

The main limitation of this work is that the results could not be extrapolated to the upper body because most of the literature exploring the effects of concurrent training targets lower-body adaptations. Besides, the low training status of most of the studies makes it also difficult to extrapolate our findings to highly trained populations. Another limitation is that not only randomized but also nonrandomized controlled trials have been included in this review. However, the methodological quality of all the included articles was assessed and all studies were considered as of good or moderate methodological quality. Methodological differences between studies such as the program design, the methods to assess muscle hypertrophy, the included sample, or the length of the intervention may have also influenced muscle hypertrophy outcomes and contributed to some of the heterogeneity in the data. For example, some studies included in the meta-analysis prescribed resistance training programs that were designed primarily to elicit improvements in maximal strength focusing on low-volume, high-intensity training (51,62). This low-volume scenario may have been suboptimal for maximizing hypertrophy (9,22,25). On the contrary, to study the hypertrophic adaptations to resistance training with or without added endurance type training, resistance training would require to be load and volume equated (not only duration) so that both groups (resistance and concurrent training) receive homogenous training stress. This was also acknowledged by Lundberg et al. (49) where the group performing concurrent training received a higher training load confusing whether the results reported were attributed to the higher overall training volume and not the added endurance exercise per se. Similarly, and despite monitoring the internal load of every subject, Fyfe et al. (22) found that the concurrent

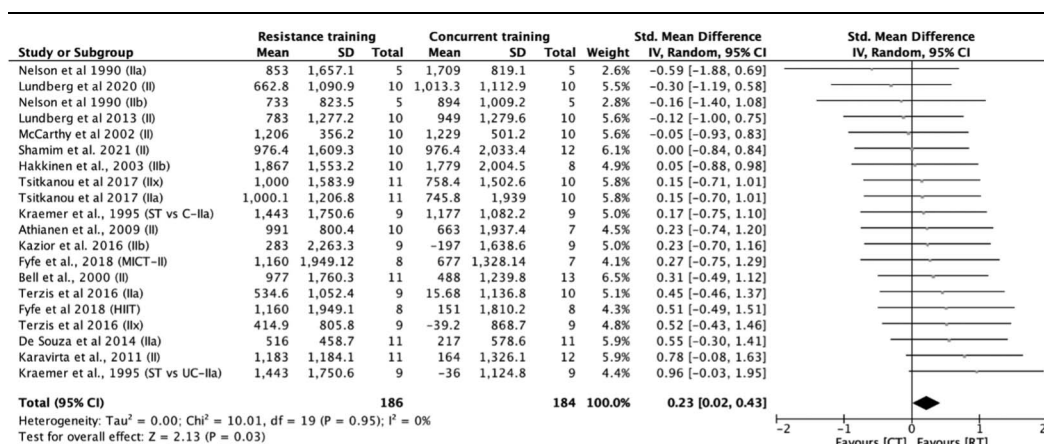


Figure 10. Type II muscle fiber hypertrophy when considering all studies.

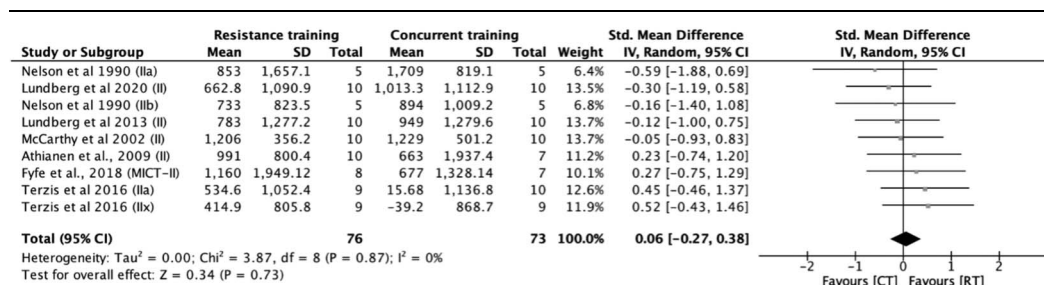


Figure 11. Type II muscle fiber hypertrophy with mixed concurrent training.

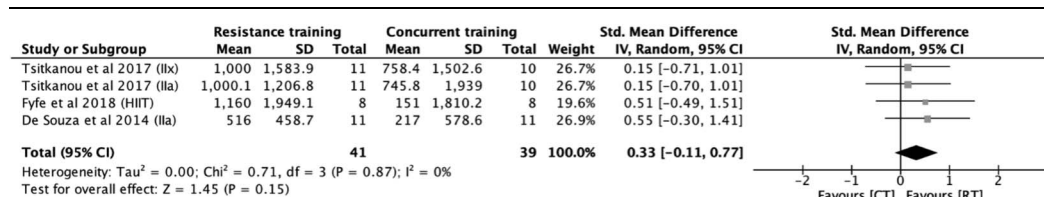


Figure 12. Type II muscle fiber hypertrophy with intermittent concurrent training.

group performing HIIT undertook a higher total internal load, suggesting that factors such as sleep quality, proper dietary and protein intake (specially around training), and levels of muscle soreness may alter the internal-external load relationship and hence all the training induced adaptations. Shamim et al. (69) also reported unbalanced conditions where recovery time between sessions was significantly less in the concurrent training group when compared with the resistance training group. Therefore, the increased fatigue and muscle soreness along with reduced recovery associated with concurrent training may have altered the subsequent RT session performance, thus accentuating the important role that nutrition, sleep quality, and overall recovery plays in this complex equation or paradigm when aiming for muscle growth. However, it should be noted that subjects had at least 24 hours of rest between resistance and aerobic-based exercise in this study (69); thus, it could minimize potential interferences in muscle hypertrophy considering that this is time enough to recover from fatigue induced by endurance training (16,22,38,45). Along these lines, Ronnestad et al. (62) compared muscle growth after 12 weeks of RT in untrained subjects and highly trained endurance athletes while they tracked the dietary intake. The endurance athletes trained an average of 10 hours per week while the untrained individuals conducted resistance training alone. The RT nonathlete group almost doubled the whole-muscle growth as the concurrently trained endurance athletes and, despite the athletes undertaking

larger volumes of training, both profiles (untrained and athletes) had the same total energy, protein, carbohydrate, and fat intakes. Although the differences in hypertrophy between groups could be solely associated to training status, it is reasonable to think that nutritional intake was suboptimal to support hypertrophy in the endurance athletes while adequate in the untrained individuals.

This systematic review and meta-analysis showed that resistance training induces greater type I and type II muscle fiber hypertrophy than concurrent training incorporating either high-intensity interval training alone or in combination with continuous endurance training, but not compared with concurrent training incorporating continuous endurance training. By contrast, at the whole-muscle level, there were no differences for hypertrophy outcomes between resistance training and concurrent training, regardless of the endurance training modality used.

Practical Applications

Athletes performing CT should choose exercise order, volume, and intensity based on individual preference or the goals of the specific training phase, warranting a cautioned interpretation of expected changes in hypertrophy, power, or strength when considering different training approaches. For example, in a power lifting context, voluminous blocks

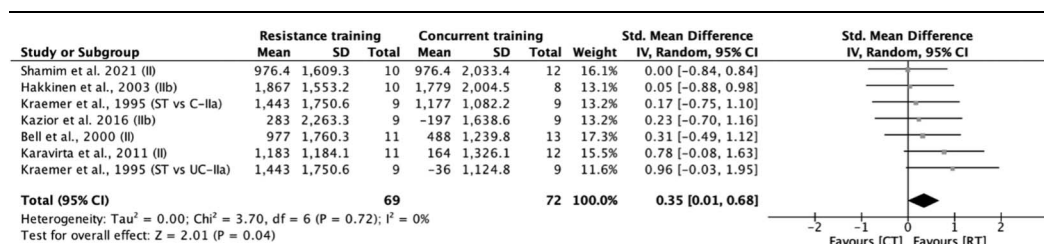


Figure 13. Type II muscle fiber hypertrophy with continuous concurrent training.

incorporating ET and RT should be positioned far away from peaking phases. Athletes seeking hypertrophic, strength, or power adaptations (i.e., bodybuilders, powerlifters, weightlifters) should prioritize resistance training before endurance training giving enough time (6–24 h) to recover between sessions. Low- or moderate-intensity endurance training are recommended if both training bouts are completed in a short window of time as HIIT might be more taxing than MICT in terms of recovery. No impact modalities of ET (i.e., cycling) are preferred over running, because there is more eccentric muscle damage and increased recovery time.

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